INSTITUT POLYTECHNIQUE DE PARIS

FUNDAMENTALS OF MICROSCOPY

Biomedical Engineering - Biomedical Data Science and Applications



RÉPUBLIQUE FRANÇAISE
Liberté
Égalité
Fraternité

ip-paris.fr

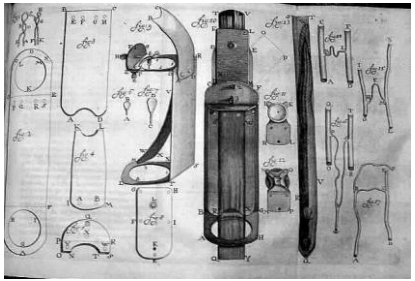
INTRODUCTION

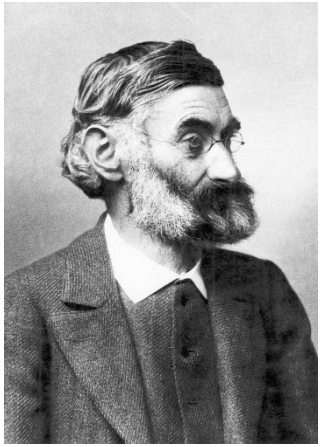


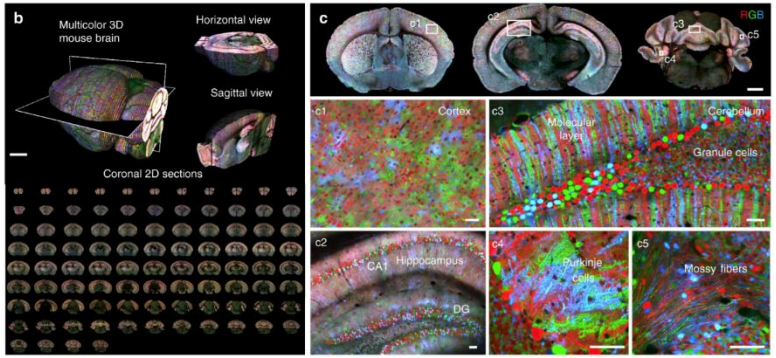




17th Century → 19th Century → Early 20th Century → Mid 20th Century → 21st Century → The age of data







INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

ORIGINS: THE BIRTH OF MICROSCOPY (17TH CENTURY)



ORIGINS: THE BIRTH OF MICROSCOPY (17TH CENTURY)

16th century | Early 17th century | Mid 17th century | Late 17th century



Hans Janssen and Zacharias Janssen (around 1590, Middelburg, Netherlands)

- Father and son spectacle makers sometimes credited with inventing the first compound microscope (10x magnification)
- Historical evidence is sparse, and the attribution comes mainly from later testimonies.

Hans Lippershey (1608, Middelburg, Netherlands)

- More firmly documented as the inventor of the telescope (patent application in 1608).
- Sometimes suggested to have made simple magnifying instruments, but this is disputed.

ORIGINS: THE BIRTH OF MICROSCOPY (17TH CENTURY)

16th century → Early 17th century → Mid 17th century → Late 17th century



Galileo Galilei (1609–1624, Italy)

- Improved the telescope (1609) and adapted it to close viewing.
- By 1624, described an occhiolino (“little eye”) which could magnify small objects

Cornelis Drebbel (1620s, England/Netherlands)

- Built compound microscopes demonstrated in London. His instruments were influential among early natural philosophers.

ORIGINS: THE BIRTH OF MICROSCOPY (17TH CENTURY)

16th century → Early 17th century → Mid 17th century → Late 17th century



Giovanni Faber (1625, Rome)

- First to use the term microscope (from Greek: “small” + “to look at”) to describe Galileo’s device.

Robert Hooke (1665, England)

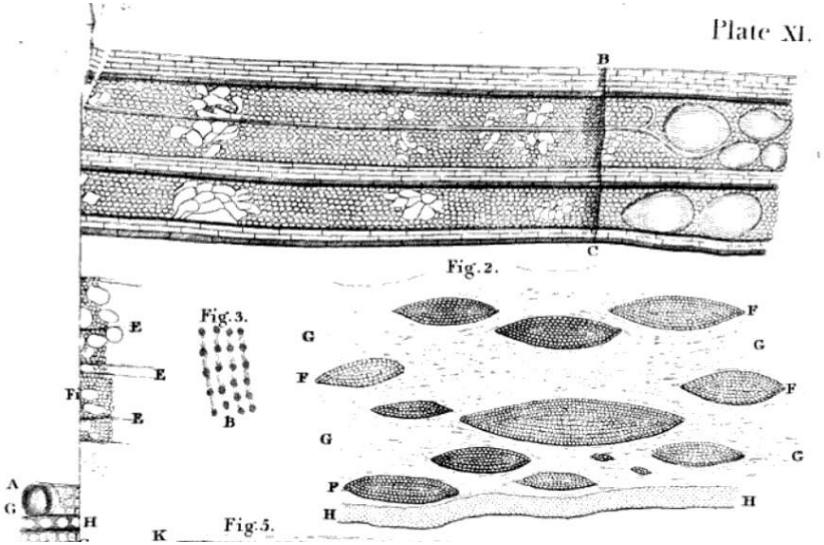
- Published Micrographia, the first major book with detailed microscope illustrations.
- Described the structure of cork, coining the term “cells”.
- Popularized the scientific potential of microscopy across Europe.

ORIGINS: THE BIRTH OF MICROSCOPY (17TH CENTURY)

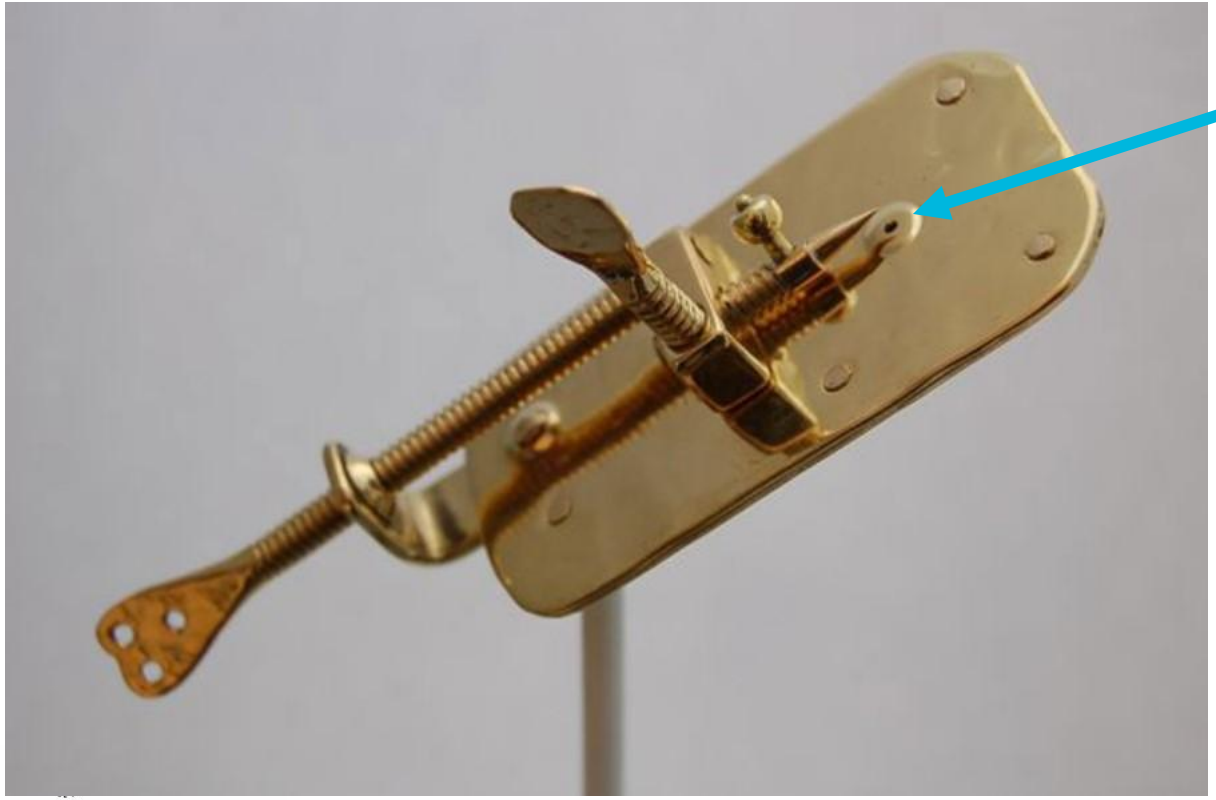
16th century → Early 17th century → Mid 17th century → **Late 17th century**

Antonie van Leeuwenhoek (1670s–1720s, Delft, Netherlands)

- Not a compound but a single-lens microscope maker, achieving up to ~200–300× magnification.
- Made first observations of bacteria (1676), sperm cells (1677), red blood cells, protozoa, muscle fibers, and more.
- Reported his findings in letters to the Royal Society (from 1673 onward).



ORIGINS: THE BIRTH OF MICROSCOPY (17TH CENTURY)



Single spherical lens

Major problem: Stigmatism

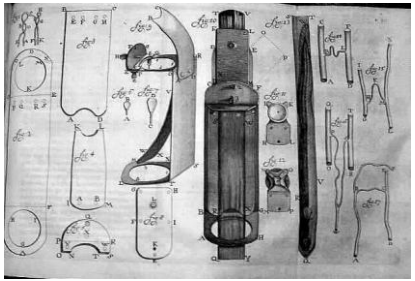
INTRODUCTION

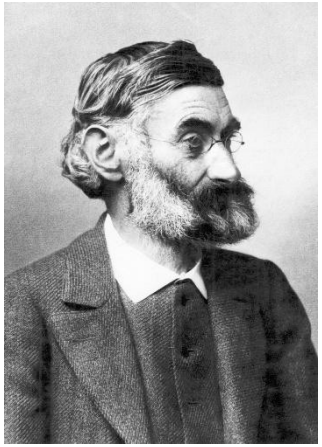


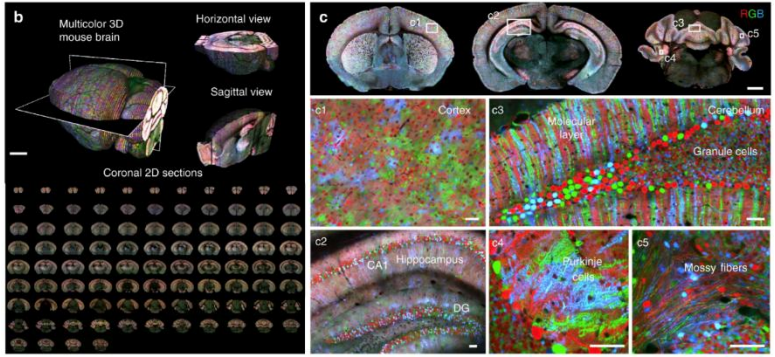




17th Century → 19th Century → Early 20th Century → Mid 20th Century → 21st Century → The age of data





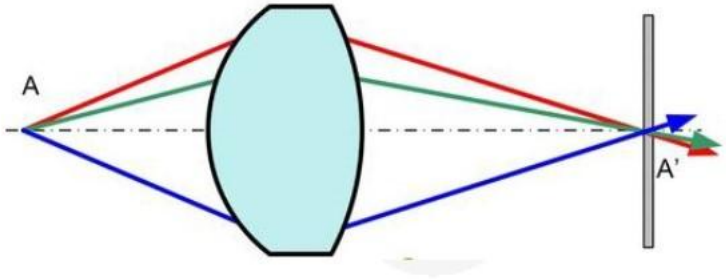


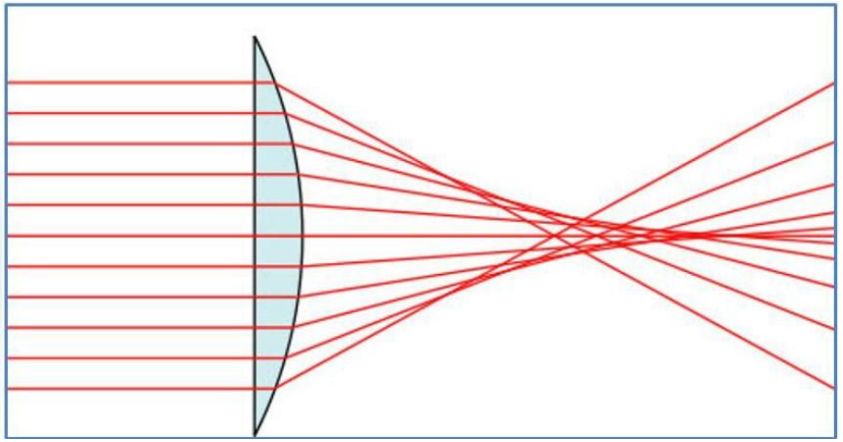
INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

Stigmatism: the image of a point is a point

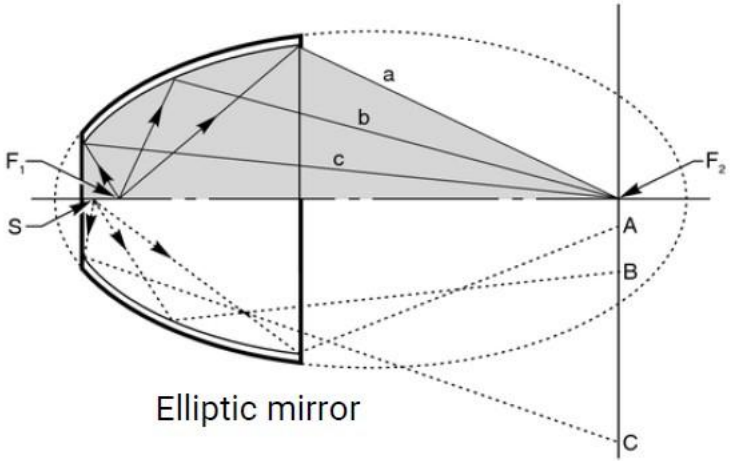
All the rays coming from A converge to A' → necessary condition for clear image





Clearly not the case for a simple lens

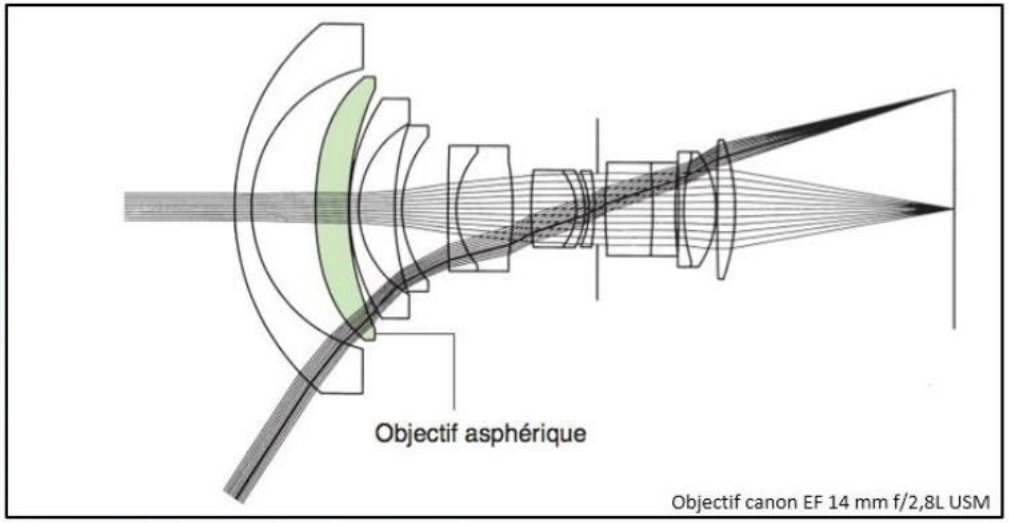
Except for a few exceptions, optical systems are not stigmatic

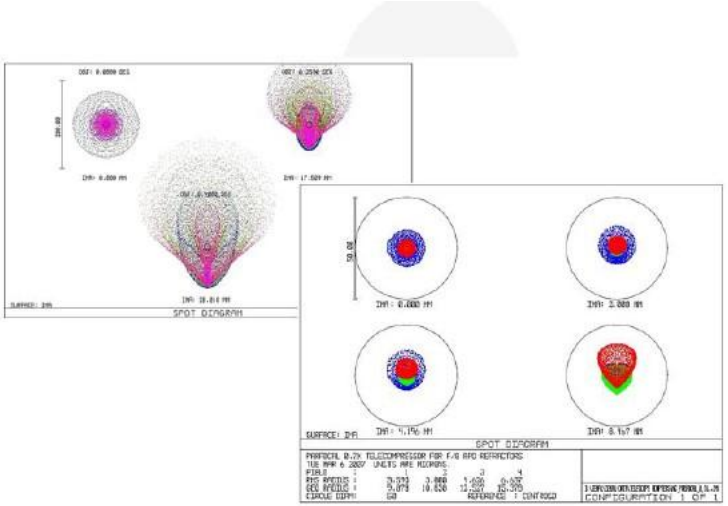


Elliptic mirror

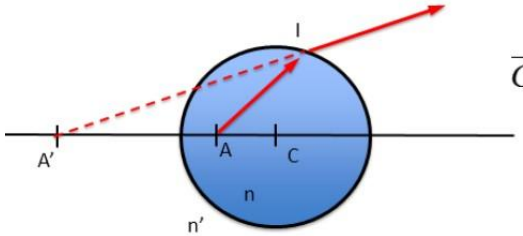
- Parabolic mirror for a point object at infinity on the optical axis
- Elliptic mirror for a conjugation between its geometric foci
- A plane mirror is always stigmatic (everypoint in space)

The goal of optical design is to reduce the spot size (Point Spread Function) to less than a pixel





Spherical diopter and Young-Weierstrass points

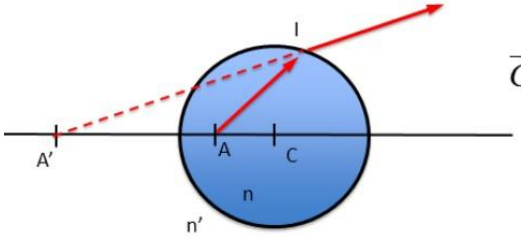


$$\overline{CA} = R\frac{n'}{n} \qquad \overline{CA'} = R\frac{n}{n'}$$

Real object/virtual image

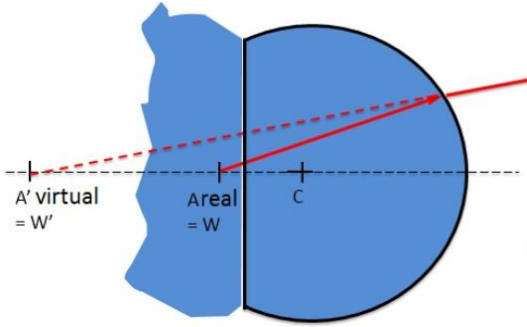
Problem: the object is inside the diopter…
Is it even useful at all?

Spherical diopter and Young-Weierstrass points



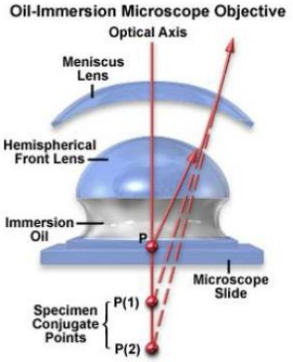
$$\overline{CA} = R\frac{n'}{n} \qquad \overline{CA'} = R\frac{n}{n'}$$

Real object/virtual image



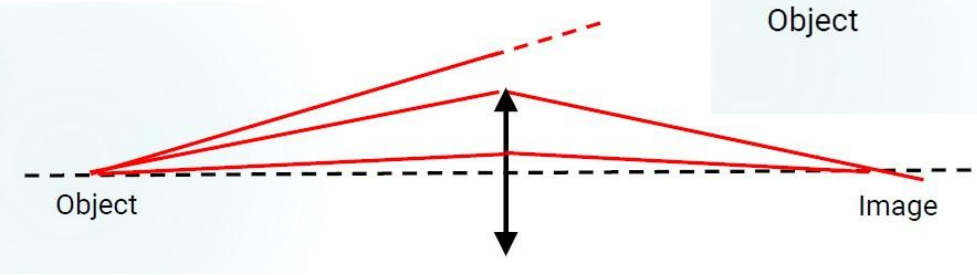
Problem: the object is inside the diopter…
Is it even useful at all?

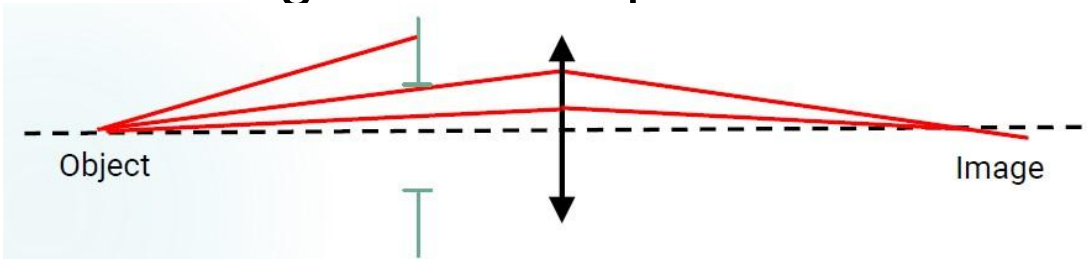
The object is put inside a liquid which has the same refractive index as glass





But if we manage to make our optical system nearly stigmatic, what is the smallest spot we can image ? → It depends on the aperture





The **pupil** (or aperture stop) is a physical element that limits the angle of rays (aperture) for an object on axis

This controls the amount of light

The image of the pupil by the optical system is called the **exit pupil** (reversly the **entrance pupil**)

For a microscope, the entrance pupil is at infinity (object space telecentric system)

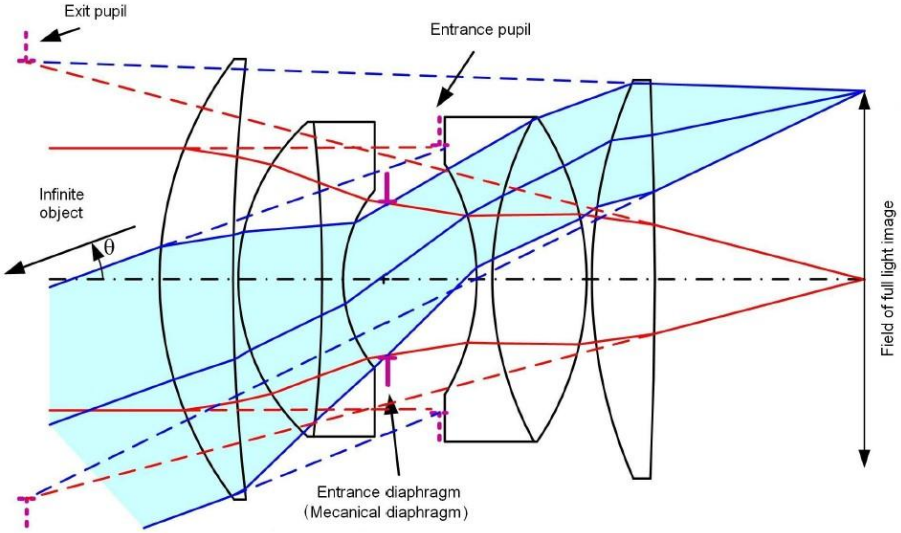
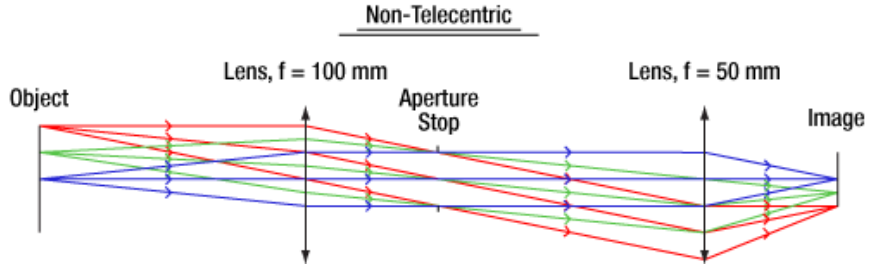


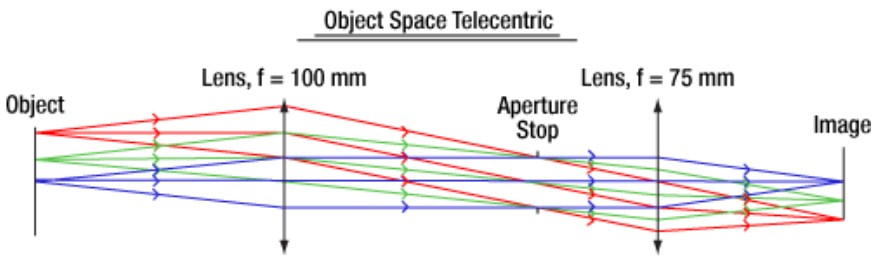
Figure 52

Object space telecentric system

- identical photometric properties regardless of position
- Identical magnification as well

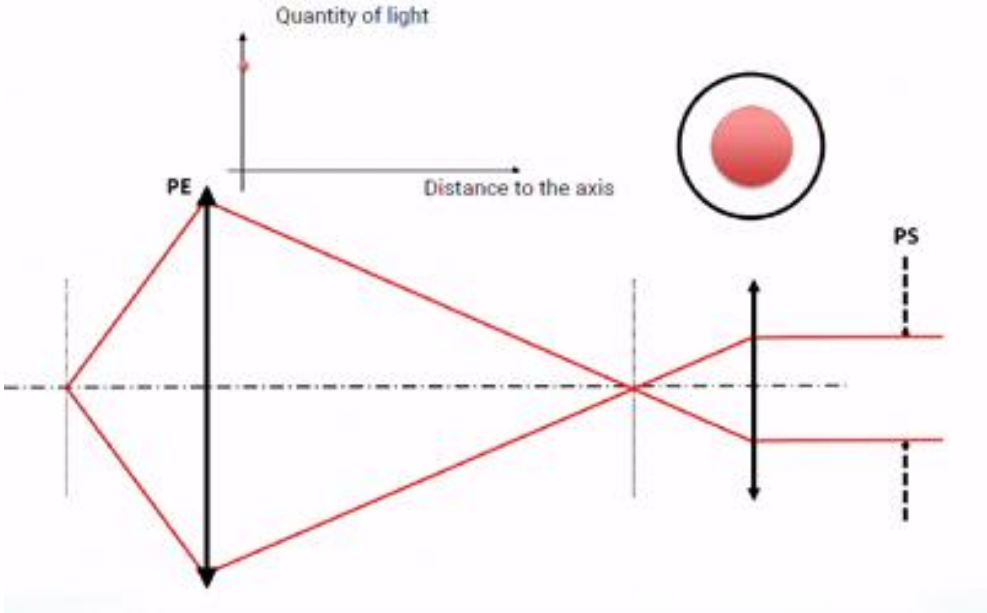




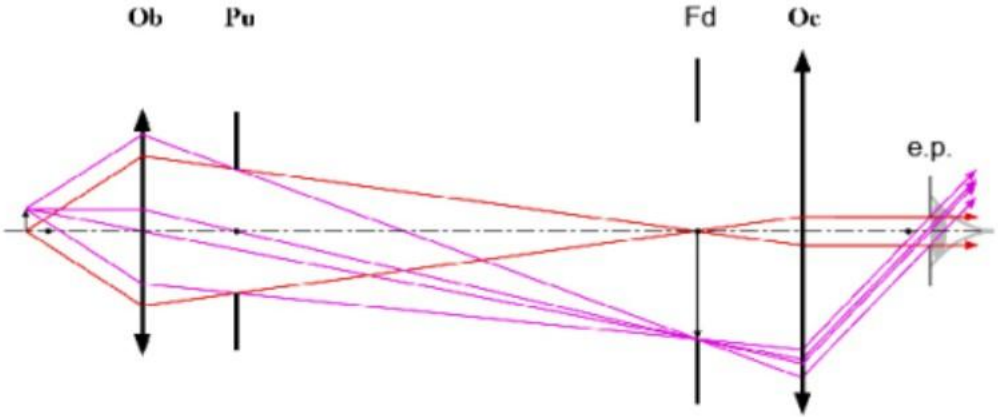




Be careful not to mix pupil and **window** (field stop)

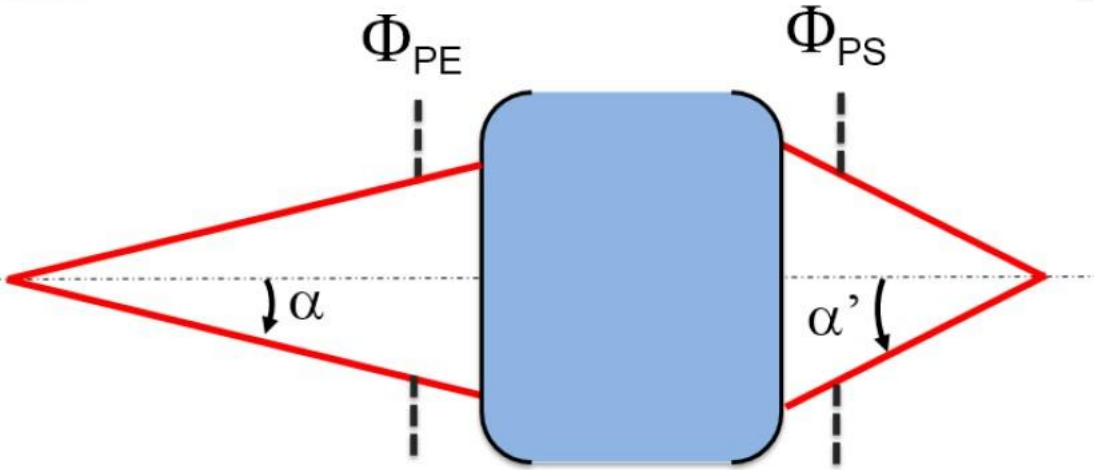


Anatomy of a simple microscope



$$P_{\text{Microscope}} = \frac{\theta'}{y} = \frac{\theta'}{y'} \times \frac{y'}{y} = P_{\text{EyePiece}} \times g_y^{\text{Objective}} = \frac{1}{f'_{\text{Microscope}}}$$

$$P_{\text{EyePiece}} = \frac{1}{f'_{\text{EyePiece}}}$$



The numerical aperture:

$$N.A._{object} = n * \sin(\alpha)$$
$$N.A._{image} = n' * \sin(\alpha')$$

The F-number (F/N):

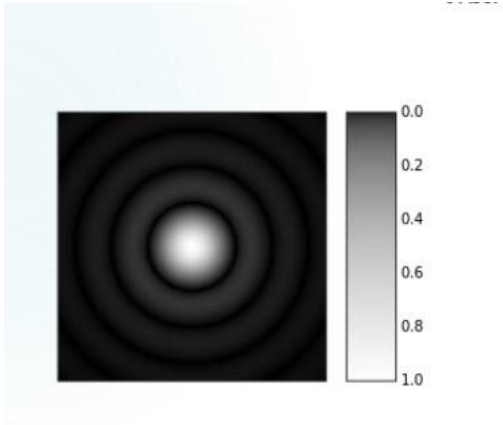
$$N = \frac{f'}{\Phi_{PS}}$$

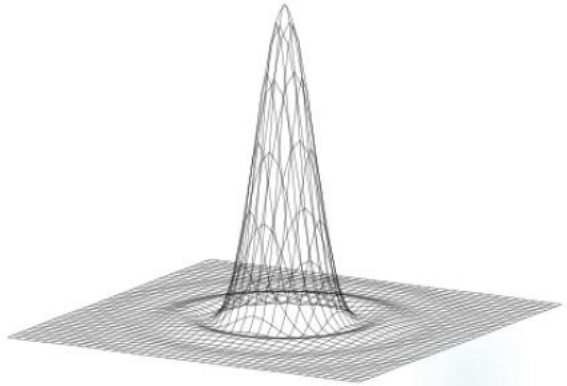
Under the best possible conditions, the optical system is limited by the diffraction of its (round) optics.

The image of a point is a spot (Point Spread Function) that takes the form of an Airy disk.

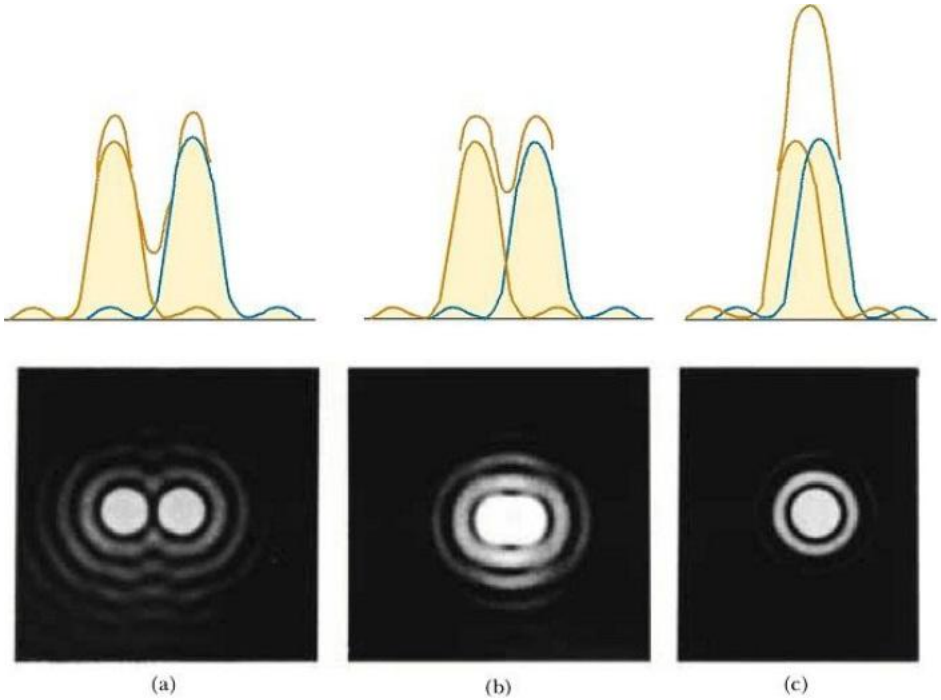
$$\emptyset = \frac{1.22\lambda}{N.A.}$$

Note: good if n>1 and large N.A. !



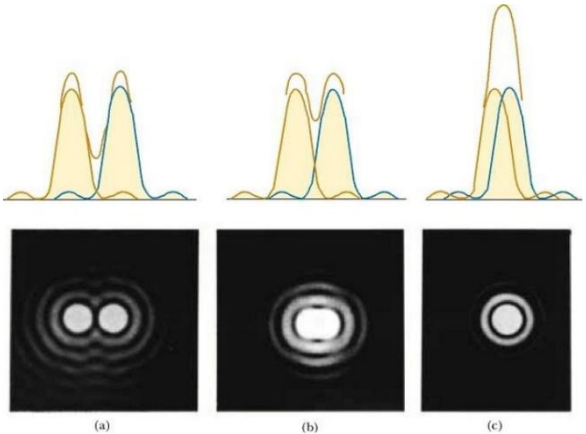


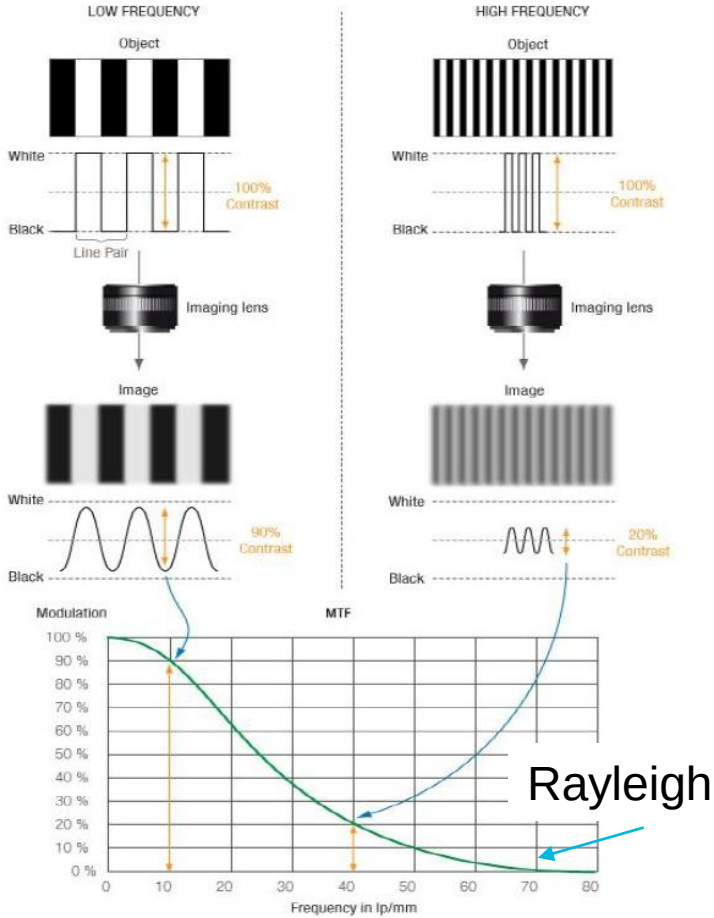
We often use Rayleigh’s criteria to quantify the resolution of a microscope



But it can also be interpreted differently as the Modulation Transfer Function

Where $MTF = \widetilde{PSF}$





19TH CENTURY: ABBE, RESOLUTION, AND THE SCIENTIFIC MICROSCOPE

An object can be properly observed only when properly illuminated.

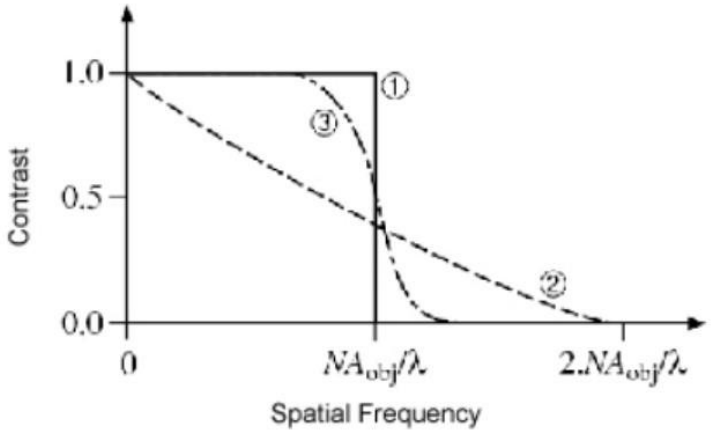
The properties of a good illumination system are the following:

- Uniformity (to observe the object and not the illumination inhomogeneities)

- An illumination area limited to the microscope field of view (to avoid stray light)

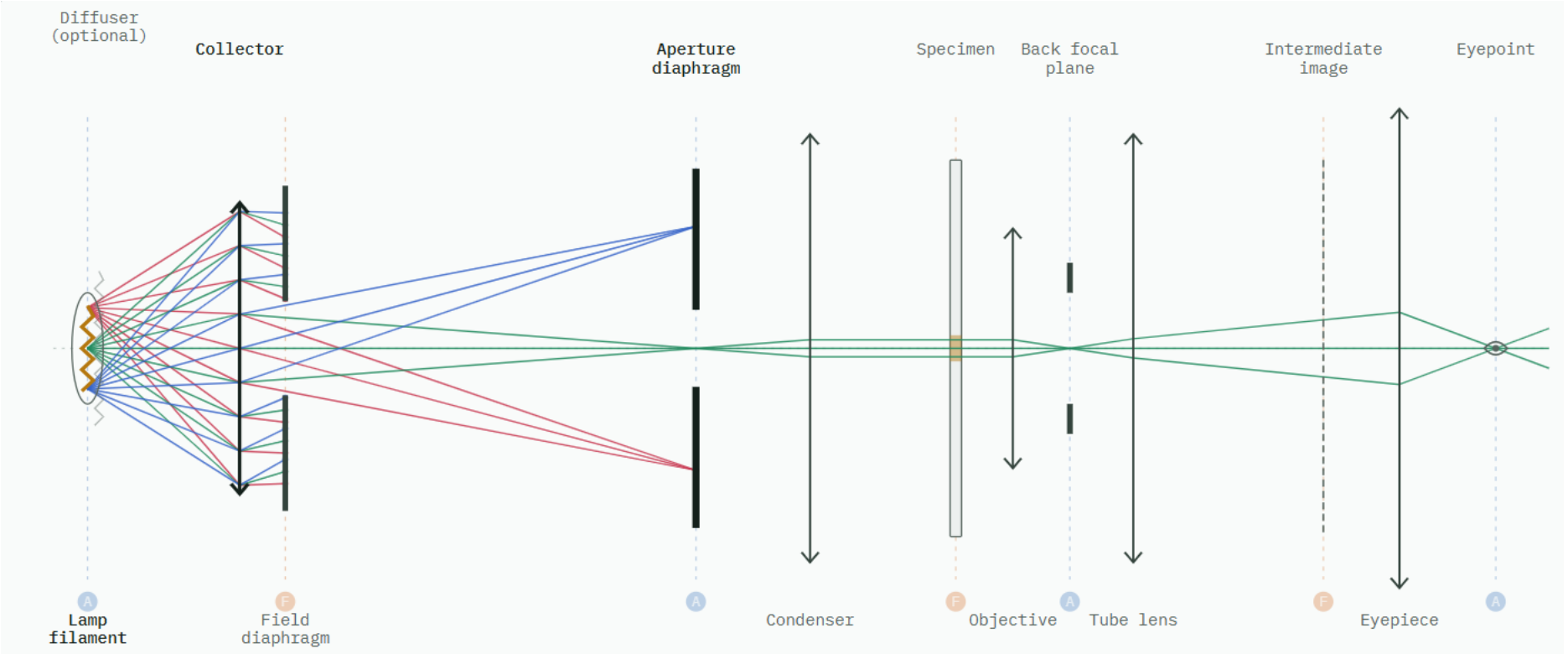
A good control of the numerical aperture (or in other words of the partial coherence of the illumination) affects the resolution

1. Coherent illumination
2. Incoherent illumination
3. Partially coherent illumination



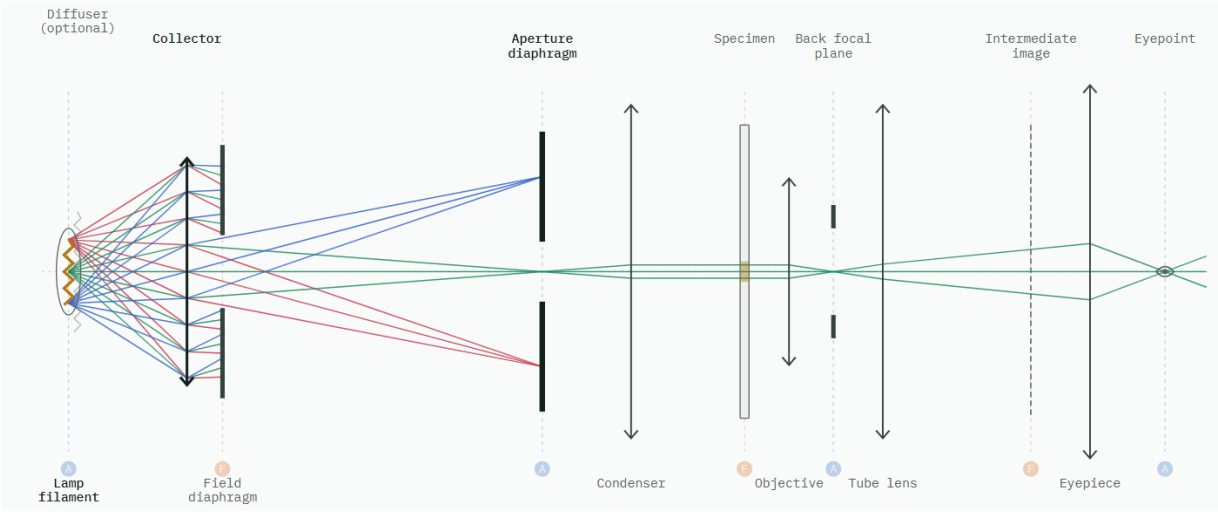
19TH CENTURY: ABBE, RESOLUTION, AND THE SCIENTIFIC MICROSCOPE

Kohler illumination:



19TH CENTURY: ABBE, RESOLUTION, AND THE SCIENTIFIC MICROSCOPE

Kohler illumination:



- The filament is sent to infinity on the sample

→ Not seen by eye

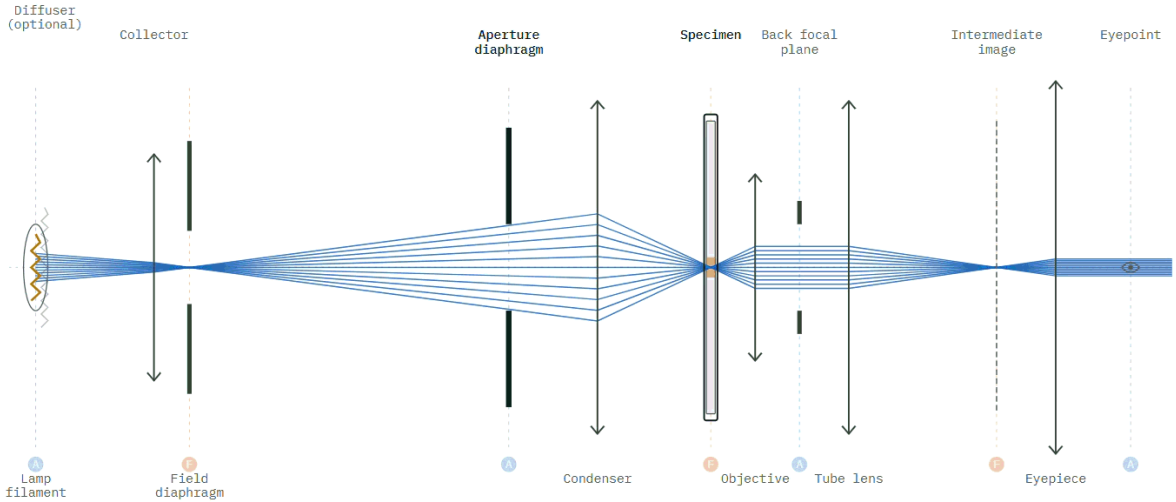
- Illumination aperture diaphragm ↔ Objective entrance pupil (**located at infinity**), via the condenser optics

→ all the points within the specimen are illuminated by the **same light cone**, which, considering that the objective is an object-space telecentric system, warrants that all the points in the microscope field of view exhibit the same photometric response

19TH CENTURY: ABBE, RESOLUTION, AND THE SCIENTIFIC MICROSCOPE

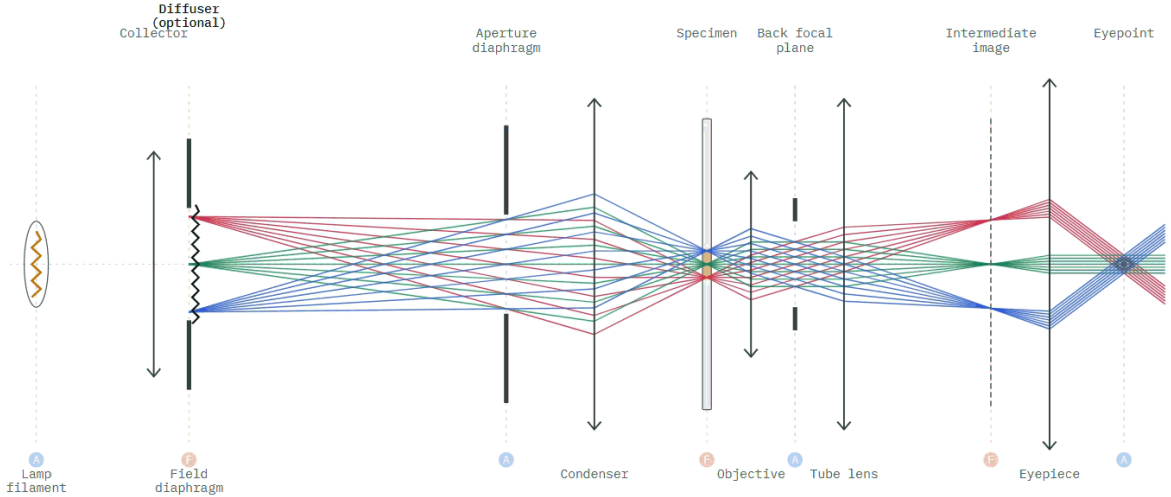
Kohler illumination:

- The aperture diaphragm can Be changed independently from The field diaphragm
→ Set the amount of light
→ Change the coherence



19TH CENTURY: ABBE, RESOLUTION, AND THE SCIENTIFIC MICROSCOPE

Kohler illumination:



- The condenser is imaged onto The sample
→ Homogeneous illumination

- The field diaphragm can be changed independently from the aperture diaphragm
→ Controls stray light and vignetting

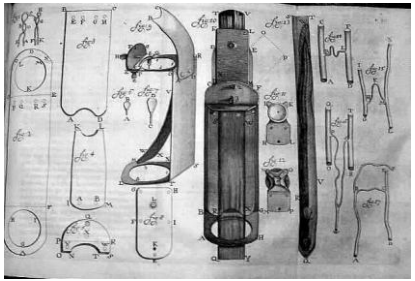
INTRODUCTION





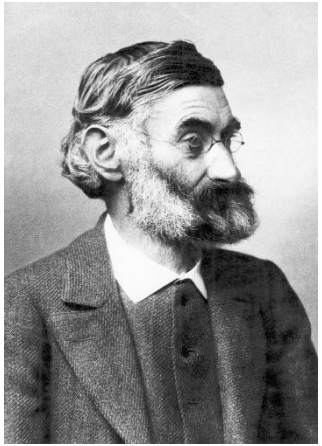


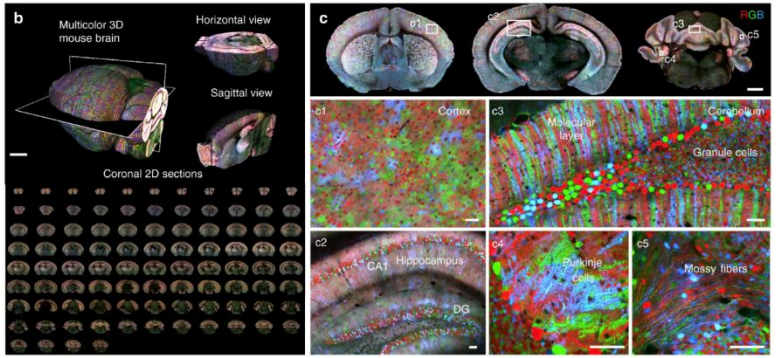
17th Century → 19th Century → Early 20th Century → Mid 20th Century → 21st Century → The age of data



INSTITUT POLYTECHNIQUE DE PARIS

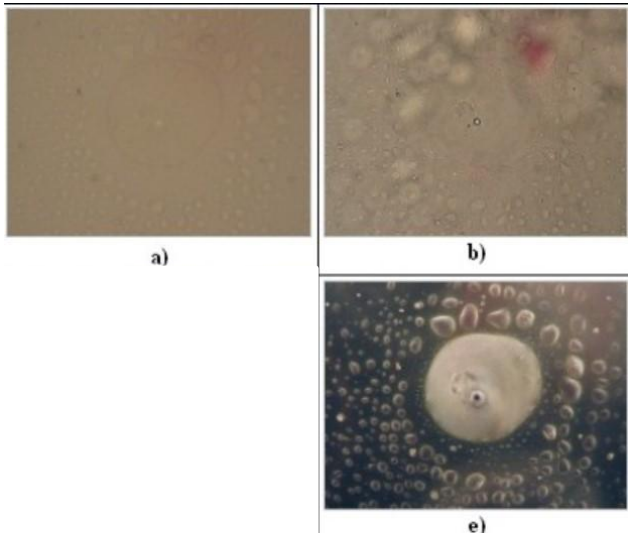
ENSTA ENSAE TELECOM Paris TELECOM SudParis





EARLY 20TH CENTURY: NEW CONTRAST MECHANISMS

- Transparent specimens are difficult to observe under a microscope since they produce smooth, low-contrast images
- Although such specimens barely change light intensity, they do modify phase or polarization, which can be converted into visible intensity variations using specialized microscope systems.
- These contrast techniques are widely available for both transmitted-light and reflected-light microscopy.



→ Dark field and phase microscopy, Nomarski's differential interference contrast

EARLY 20TH CENTURY: NEW CONTRAST MECHANISMS



Zernike's phase contrast microscopy:

Let's consider a transparent object and its complex transmittance t(M) where M represents a dot on the preparation plane:

$$t(M) = \exp(i.\Phi(M))$$

where Φ is a real value function representing the phase difference introduced by the object at each point of the object. The global phase of the wave having no importance, we choose mean=0.

If the phase variations are weak: $t(M) \approx 1 + i.\Phi(M)$

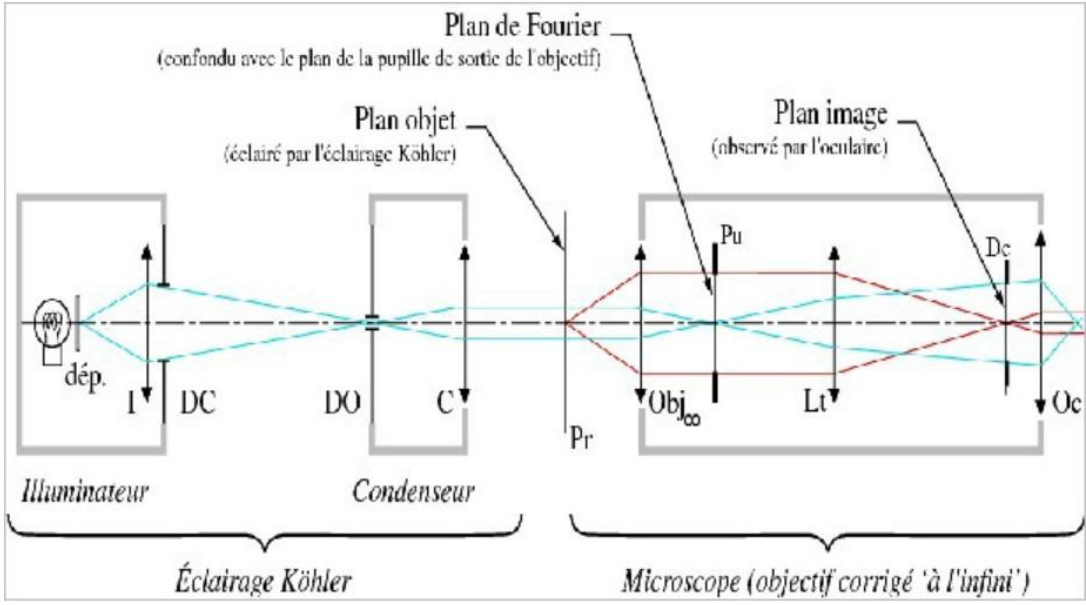
Is this what our eyes observe?

Phase contrast microscopy:

The objective lens performs a double Fourier transform:
$t(M) \approx 1 + i.\Phi(M) \rightarrow \delta + i.\widetilde{\Phi} \rightarrow 1 + i.\Phi(-M)$ and the brightness seen with our eyes is uniform, close to 1.

What if we could remove the δ ?

Do we have access to the Fourier plane ? → Yes !



EARLY 20TH CENTURY: NEW CONTRAST MECHANISMS





Phase contrast microscopy:

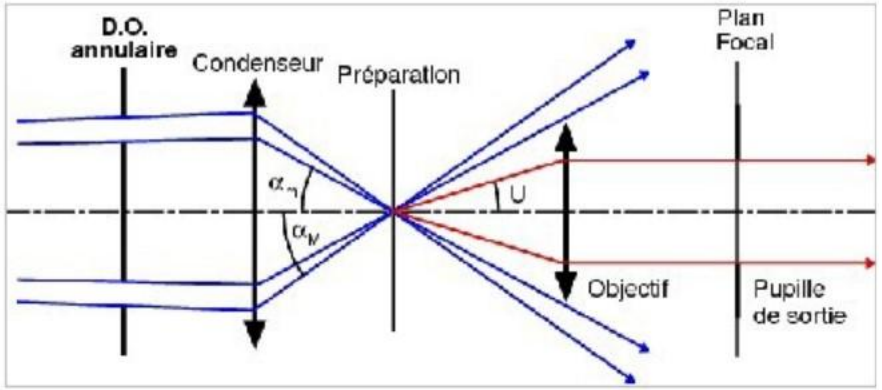
Glass slide in the Fourier place to induce a $\frac{\pi}{2}$ phase shift and a central attenuation :

$$t \approx 1 + i.\Phi \xrightarrow{TF\ 1} \pm \mathbf{i}.\boldsymbol{\gamma}.\delta + i.\tilde{\Phi} \xrightarrow{TF\ 2} \pm \mathbf{i}.\boldsymbol{\gamma} + i.\tilde{\tilde{\Phi}} = i\left(\pm\gamma + \check{\Phi}\right)$$

$$\left(\pm\gamma + \check{\Phi}\right)^2 \approx \gamma^2 \pm 2\gamma\check{\Phi} \approx \gamma^2\left(1 \pm \frac{2\check{\Phi}}{\gamma}\right)$$

$$C = \frac{\Delta\Phi}{\gamma}$$

Schlieren (Dark field): Annular illumination letting only scattered light through the objective



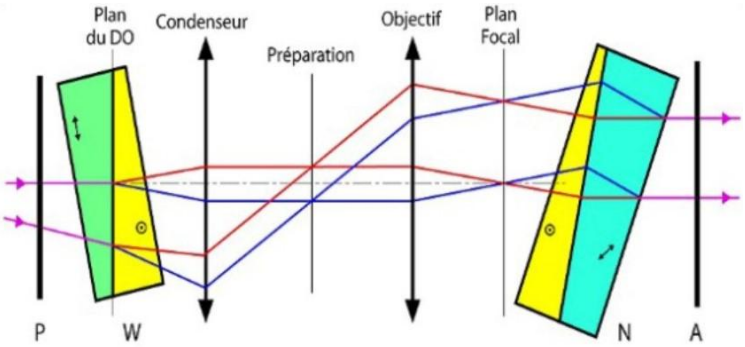
INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

EARLY 20TH CENTURY: NEW CONTRAST MECHANISMS

Normarski's differential interference contrast:

- Based on interference of polarized light.
- A polarizer produces linearly polarized light, which is then split into two beams with slightly different paths (sheared) using a Nomarski-modified Wollaston prism.



- These beams pass through adjacent regions of the specimen, acquiring different phase shifts depending on local variations in optical path length (refractive index × thickness).
- After recombination by a second prism and analyzer, the phase differences are converted into intensity differences, producing an image with apparent three-dimensional relief.

EARLY 20TH CENTURY: NEW CONTRAST MECHANISMS

Fourier Ptychographic Microscopy

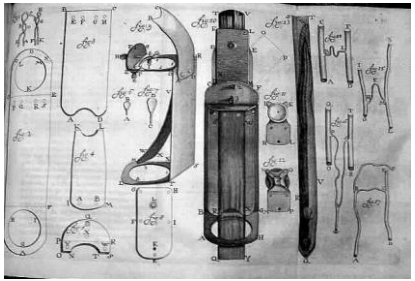
INTRODUCTION

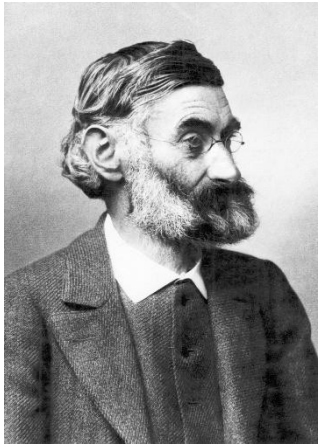


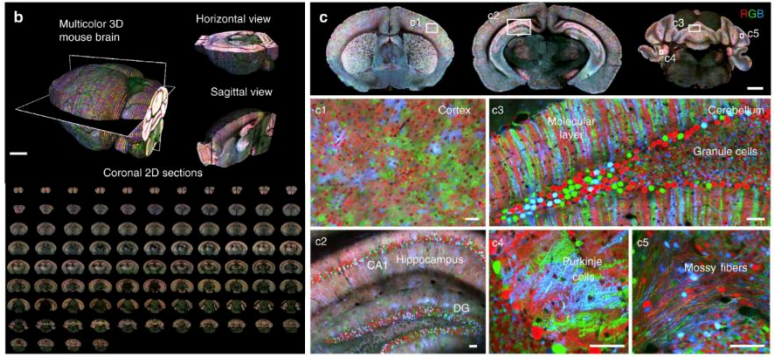




17th Century → 19th Century → Early 20th Century → Mid 20th Century → 21tst Century → The age of data



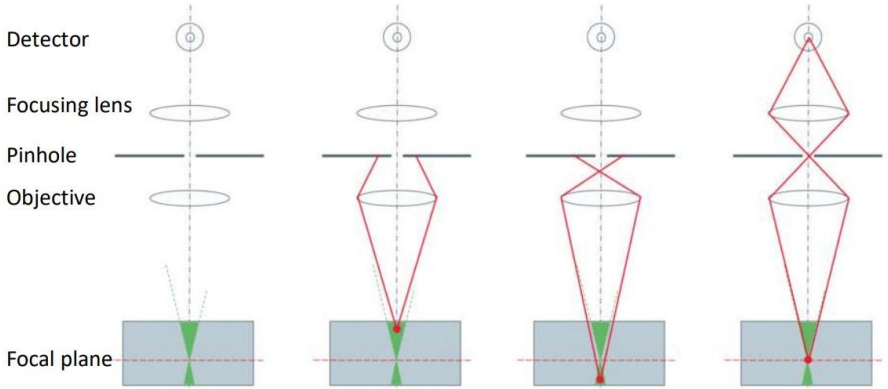


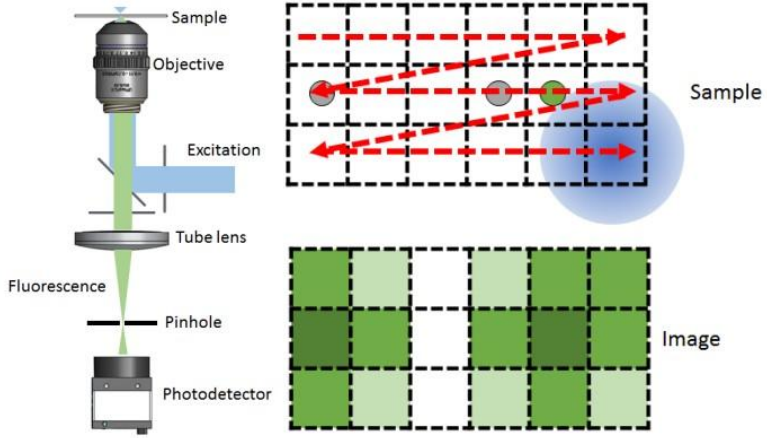


INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

Fluorescence and confocal microscopy

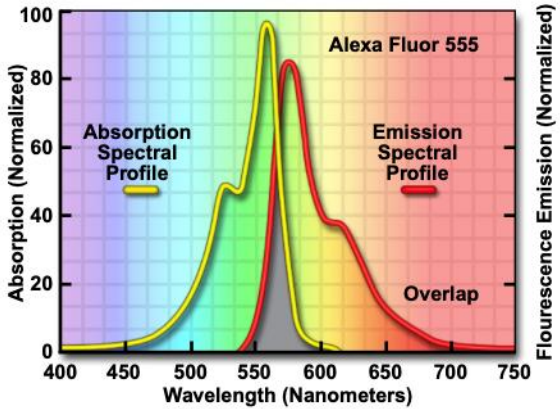


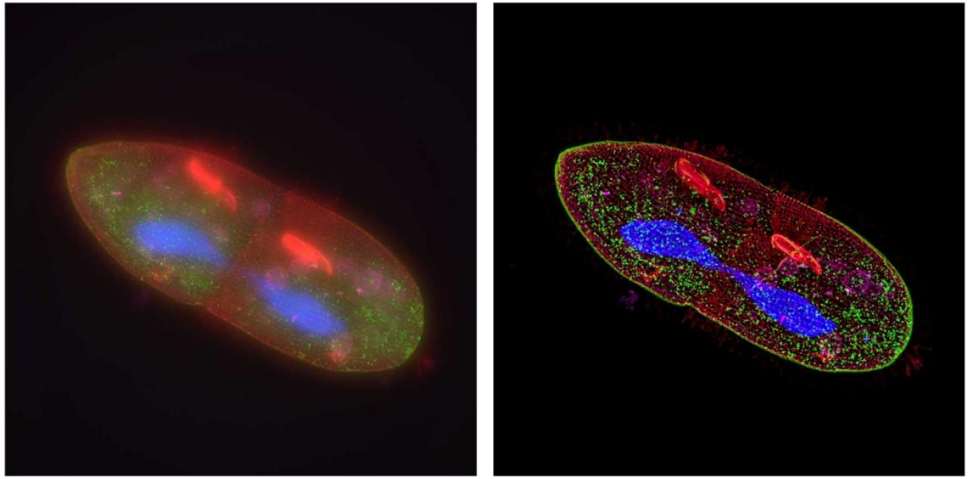


Spot illumination coupled to a scanning mechanism (sample stage, objective)

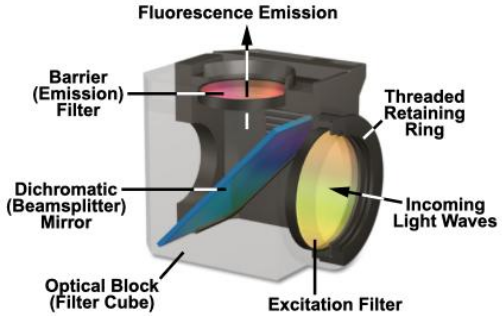
Fluorescence and confocal microscopy

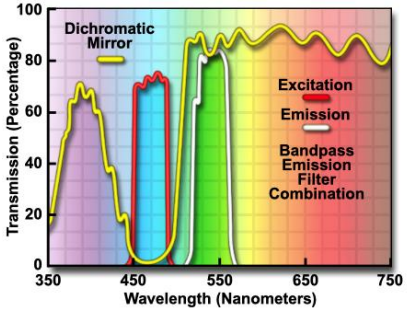
Addition of fluorescent tags
Pros: targeted, high signal
Cons: bio-chemistry

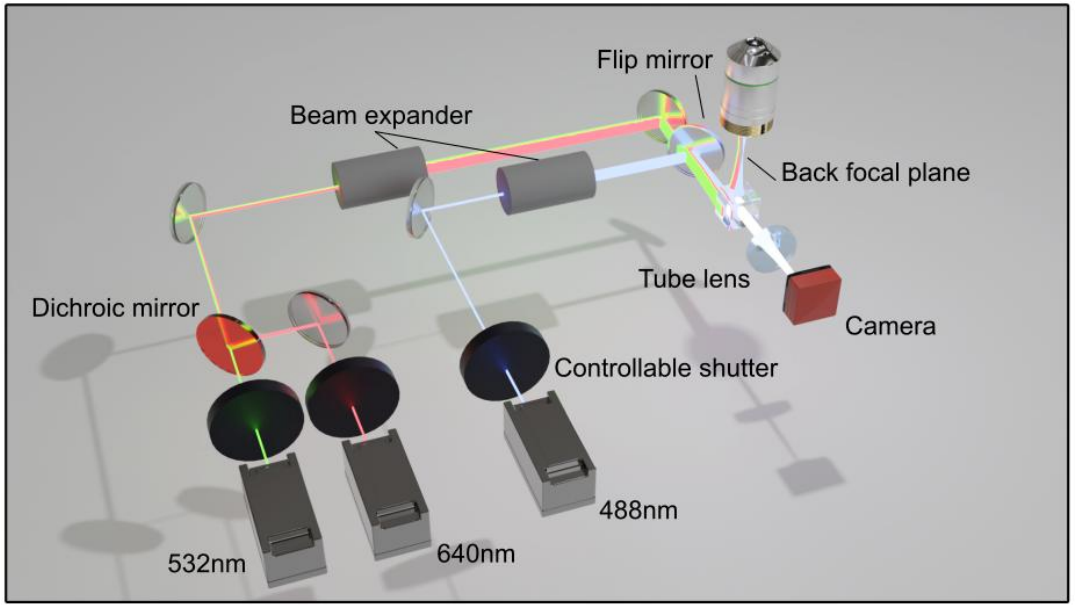




Fluorescence and confocal microscopy







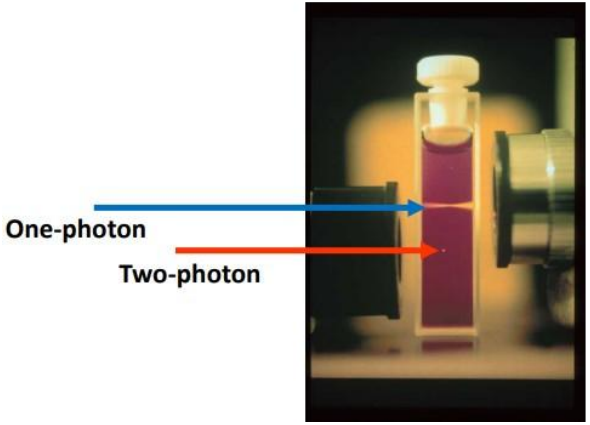
INSTITUT POLYTECHNIQUE DE PARIS

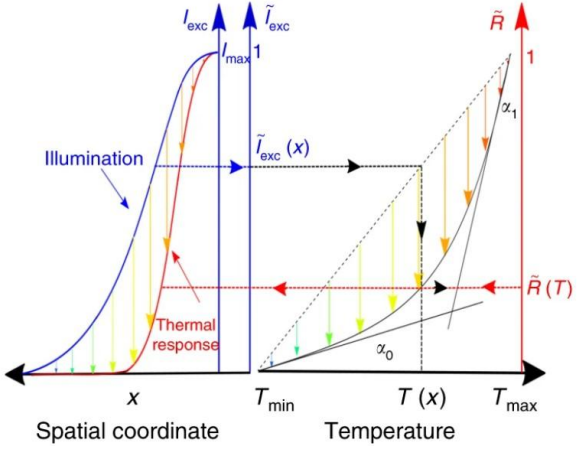
ENSTA ENSAE TELECOM Paris TELECOM SudParis

Beyond confocal:

Using non-linear phenomena to achieve smaller excitation volume

Two-photons, three-photons, Black body radiation etc

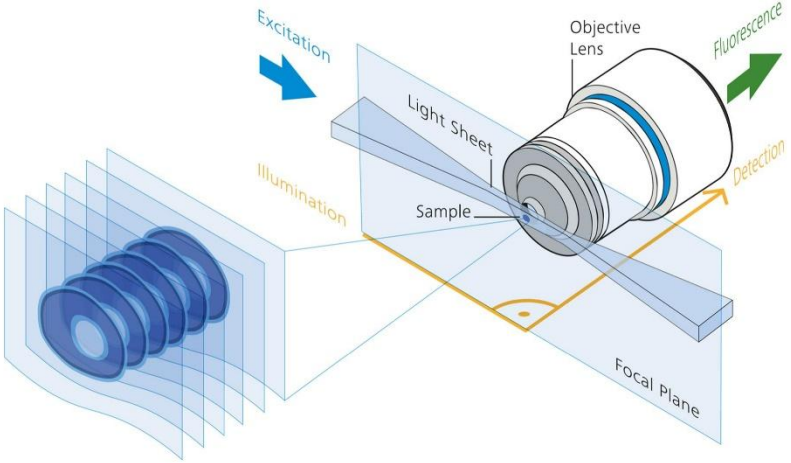


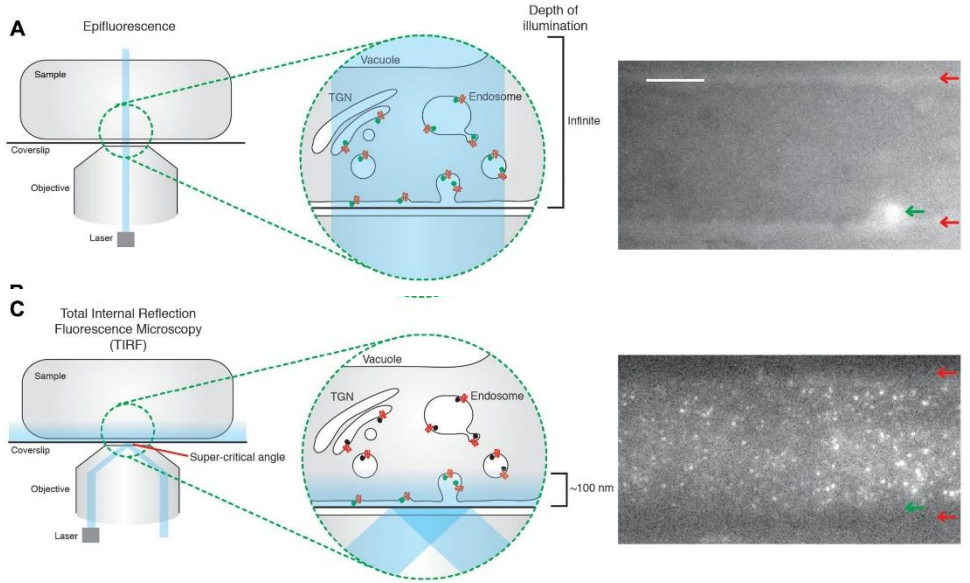


Beyond confocal:

Using inventive illumination patterns

Light sheet, structured illumination, Total Internal Reflection





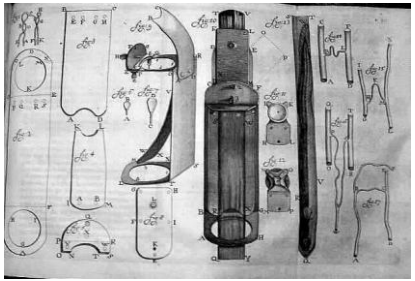
INTRODUCTION





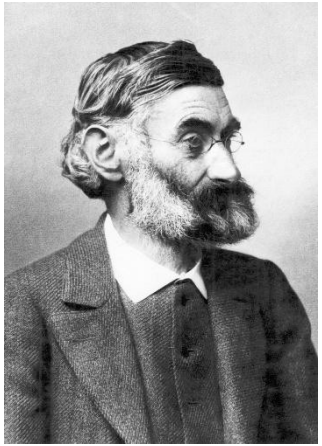


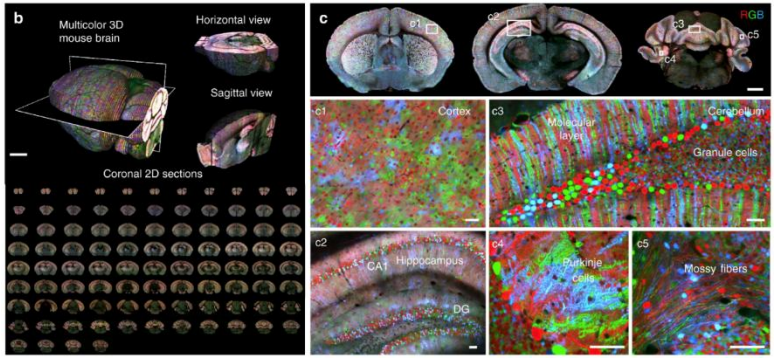
17th Century → 19th Century → Early 20th Century → Mid 20th Century → 21st Century → The age of data

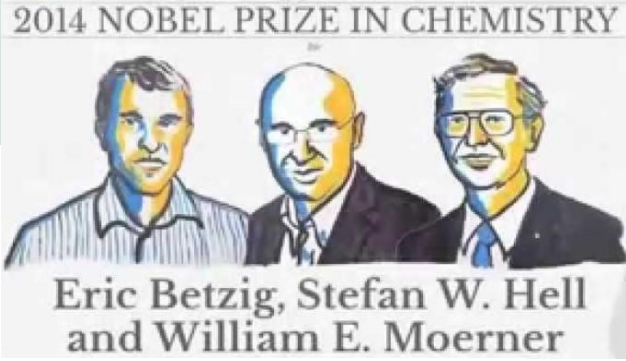


INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

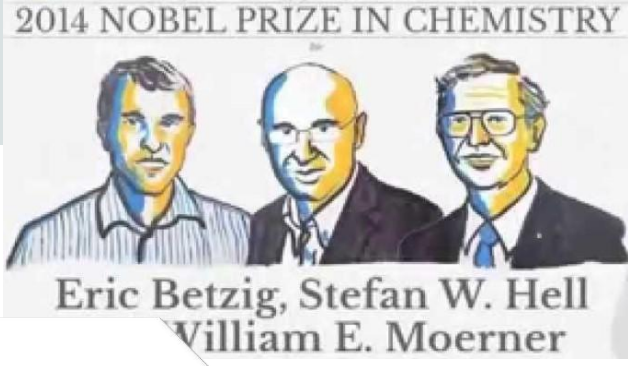






STED (Hell, 1994)

•Stefan W. Hell, a German-Romanian physicist, faced skepticism in Germany after his doctorate for attempting to surpass the Abbe limit.
•He moved to the University of Turku, Finland, where he found support to pursue his unconventional ideas.
•In 1993, inspired by quantum optics, he conceived the principle behind STED (stimulated emission depletion) microscopy.
•His 2000 breakthrough paper and subsequent appointment as director at the Max Planck Institute in 2002 enabled him to refine STED into a powerful and widely adopted imaging technique.



STED (Hell, 1994)

780 OPTICS LETTERS / Vol. 19, No. 11 / June 1, 1994

Breaking the diffraction resolution limit by stimulated emission: stimulated-emission-depletion fluorescence microscopy

Stefan W. Hell and Jan Wichmann

Department of Medical Physics, University of Turku, Tykistökatu 6, 20521 Turku, Finland

Received March 7, 1994

We propose a new type of scanning fluorescence microscope capable of resolving 35 nm in the far field. We overcome the diffraction resolution limit by employing stimulated emission to inhibit the fluorescence process in the outer regions of the excitation point-spread function. In contrast to near-field scanning optical microscopy, this method can produce three-dimensional images of translucent specimens.

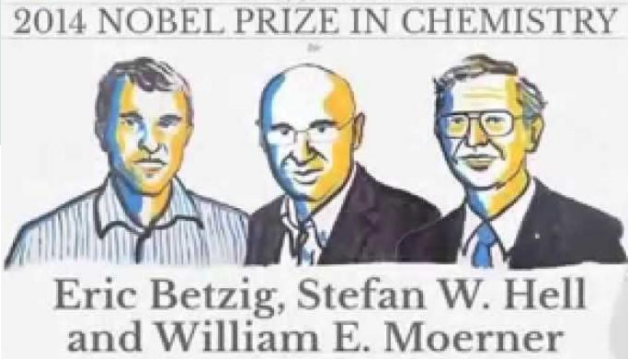
- •Stefan W ermany after his
- •He mov to pursue his unco
- •In 1993 d STED (stimulat
- •His 200 r at the Max Pla ul and widely ad

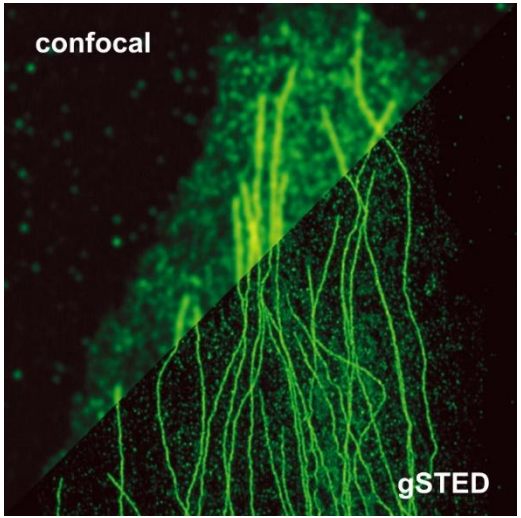
INSTITUT POLYTECHNIQUE DE PARIS

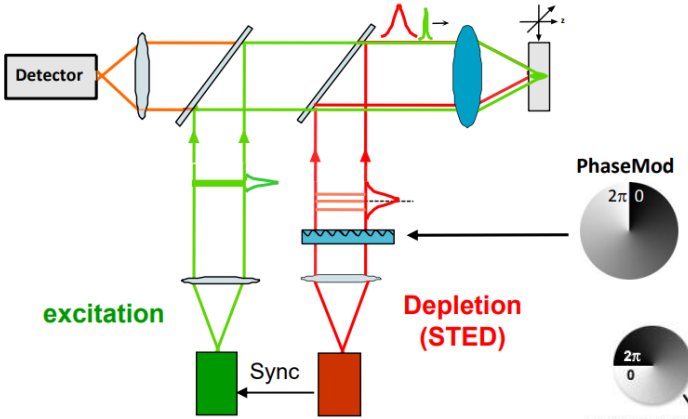
ENSTA ENSAE TELECOM Paris TELECOM SudParis

STED (Hell, 1994)

stimulated emission-depletion fluorescence microscopy







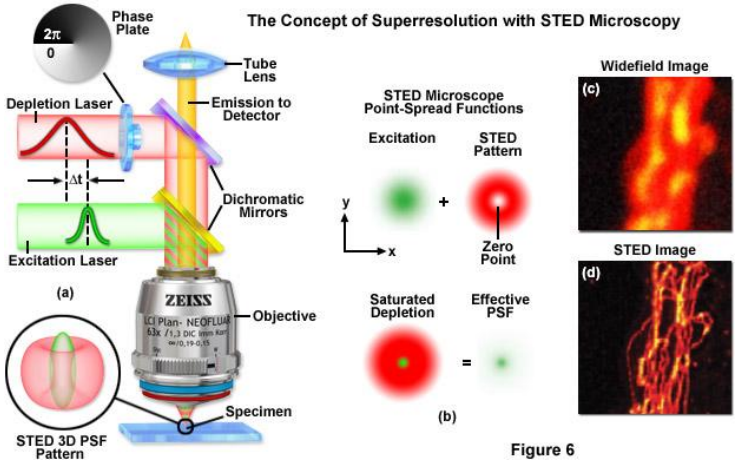


Figure 6

INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis



PALM/STORM (Betzig, Moerner, 2006)

•Eric Betzig, formerly at Bell Labs, left science disillusioned but later returned, driven by the persistent challenge of surpassing the Abbe limit.
•In 2005, he realized that switching fluorescent molecules on and off and combining many images could achieve ultra-high resolution.
•His breakthrough built on William Moerner's 1997 work at UC San Diego, where Moerner first controlled individual fluorescent molecules and manipulated green fluorescent proteins.
•Betzig's concept led to a practical imaging tool at the Howard Hughes Medical Institute's Janelia Farm campus.
•In 2014, the Nobel Prize in Chemistry was awarded to Hell, Betzig, and Moerner for founding nanoscopy, enabling unprecedented imaging of proteins, organelles, and single molecules in living cells.



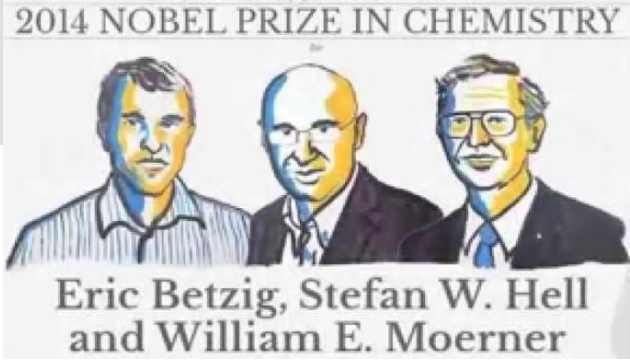
PALM/STORM (Betzig, Moerner, 2006)

Imaging Intracellular Fluorescent Proteins at Nanometer Resolution

Eric Betzig,[1,2]*† George H. Patterson,[3] Rachid Sougrat,[3] O. Wolf Lindwasser,[3] Scott Olenych,[4] Juan S. Bonifacino,[3] Michael W. Davidson,[4] Jennifer Lippincott-Schwartz,[3] Harald F. Hess[5]*

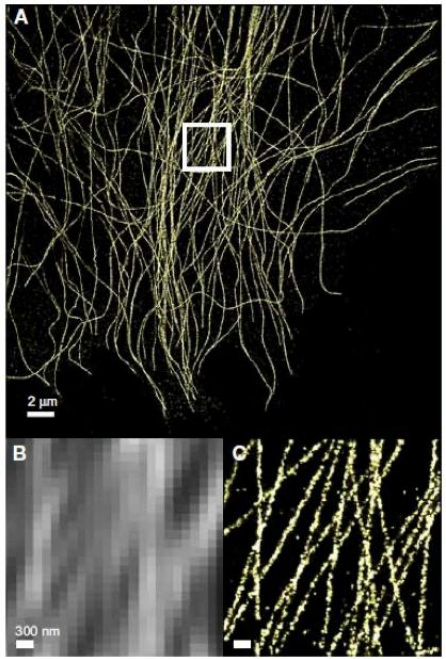
We introduce a method for optically imaging intracellular proteins at nanometer spatial resolution. Numerous sparse subsets of photoactivatable fluorescent protein molecules were activated, localized (to ~2 to 25 nanometers), and then bleached. The aggregate position information from all subsets was then assembled into a superresolution image. We used this method—termed photoactivated localization microscopy—to image specific target proteins in thin sections of lysosomes and mitochondria; in fixed whole cells, we imaged vinculin at focal adhesions, actin in a lamellipodium, and the distribution of the retroviral protein Gag at the plasma membrane.

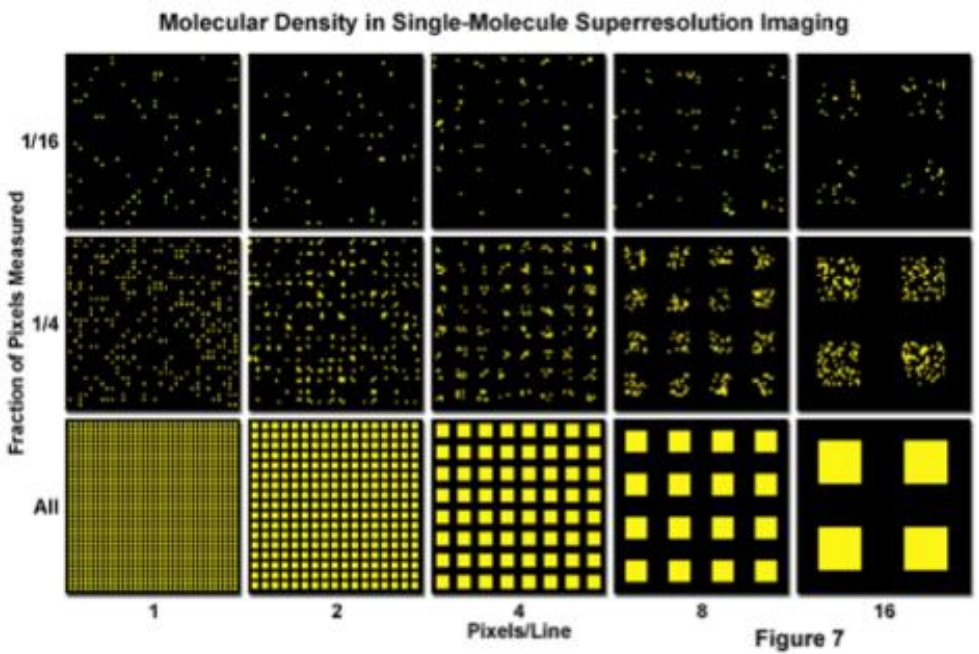
- Eric Betzig … urned, driven by the persi…
- In 2005, he … l combining many image…
- His breakth… o, where Moerner firs… d green fluorescent …
- Betzig's co… Medical Institute's Jane…
- In 2014, the Nobel Prize in Chemistry was awarded to Hell, Betzig, and Moerner for founding nanoscopy, enabling unprecedented imaging of proteins, organelles, and single molecules in living cells.



PALM/STORM (Betzig, Moerner, 2006)

Stochastic Optical Reconstruction Microscopy
Photoactivated Localization Microscopy





INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

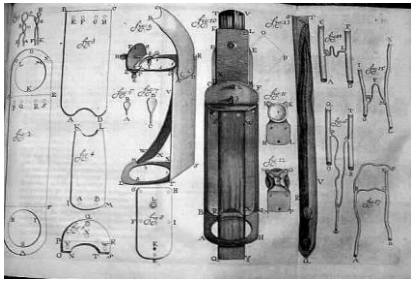
INTRODUCTION

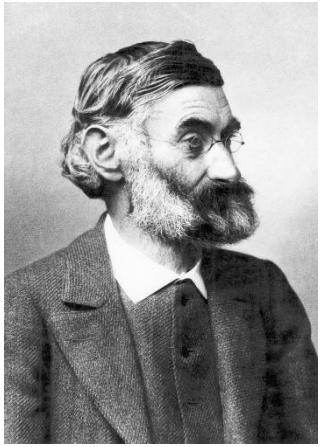


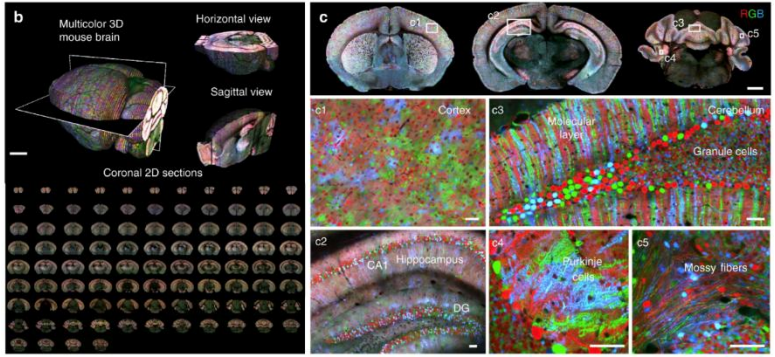




17th Century → 19th Century → Early 20th Century → Mid 20th Century → 21st Century → The age of data





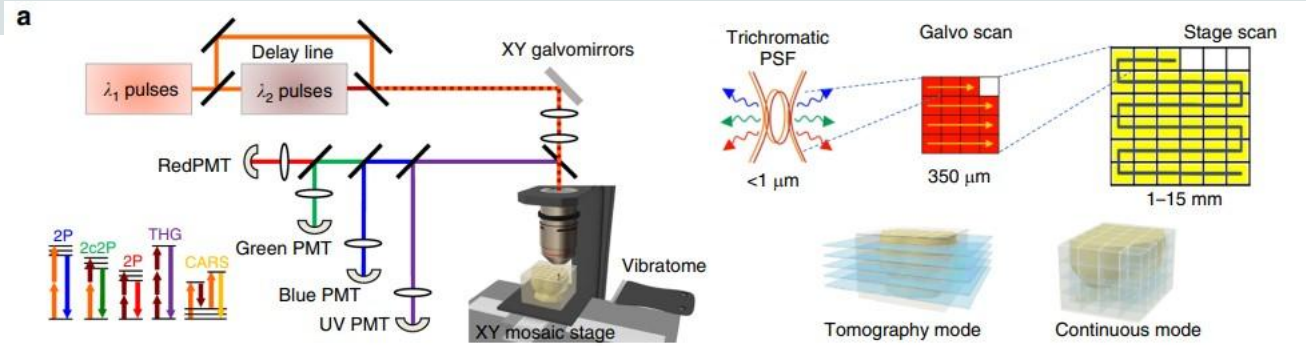


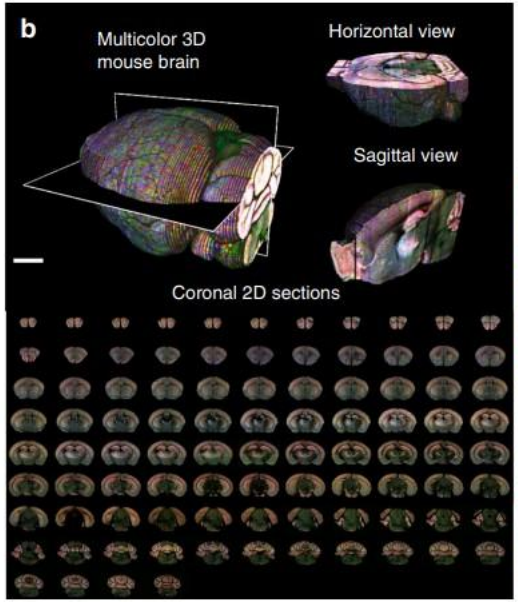
INSTITUT POLYTECHNIQUE DE PARIS

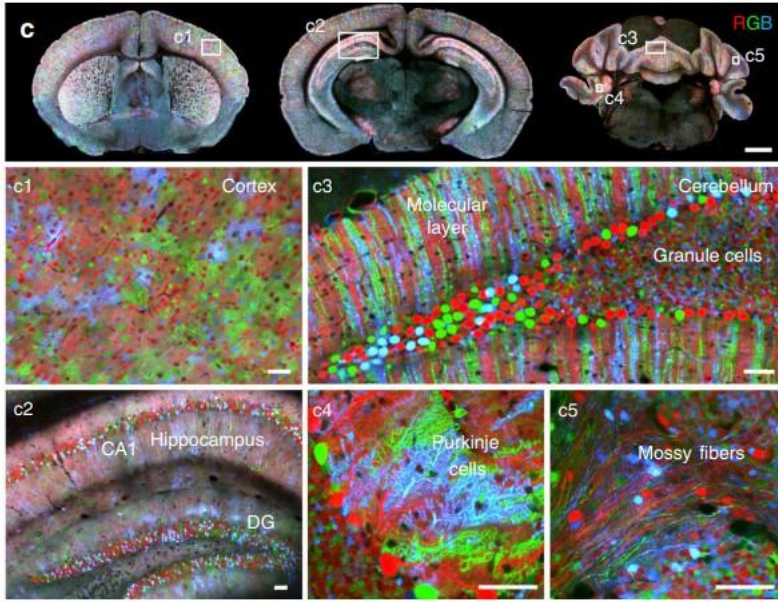
ENSTA ENSAE TELECOM Paris TELECOM SudParis

THE AGE OF DATA: WORK DONE AT THE LOB

- brainbow tissue imaging
- fluorescence (2P, 3P Two/Three-Photon Fluorescence)
- harmonic generation (SHG, THG Secon/Third Harmonic Generation)
- wave-mixing (TSFG Three-wave Sum Frequency Generation, FWM Four-Wave Mixing, CARS Coherent Anti-Stokes Raman Scattering)
- FLIM fluorescence lifetime







INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis

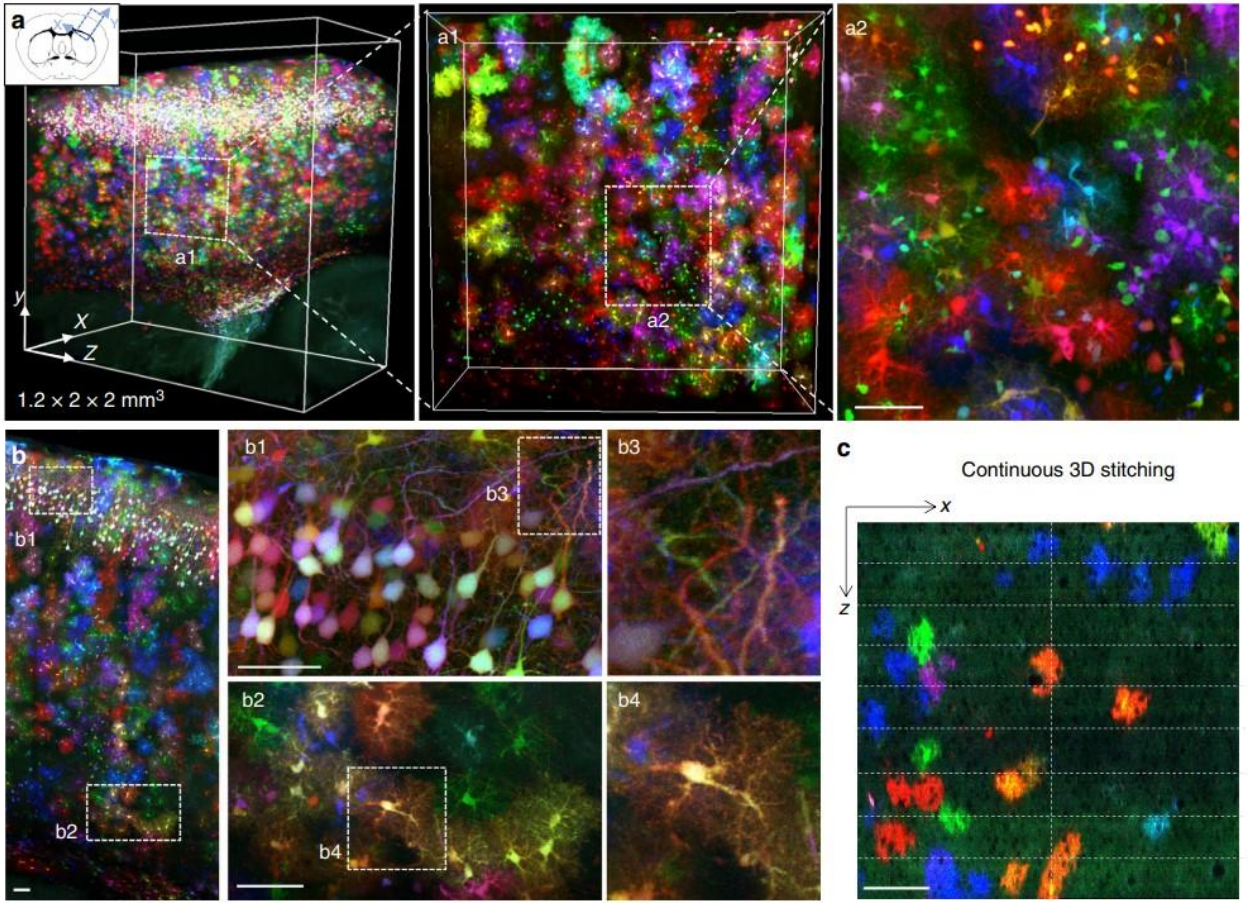
THE AGE OF DATA: WORK DONE AT THE LOB

- 1 night acquisition time
- TERABYTES of DATA

→ Months of processing

(Stitching, segmentation, localisation etc)

→ Optics lab hire expert data analysts
→ Use of Machine Learning



Computational microscopy: Fourier Ptychography: using physics-inspired deep-learning to reconstruct high resolution object from images (both intensity and phase, marker free)

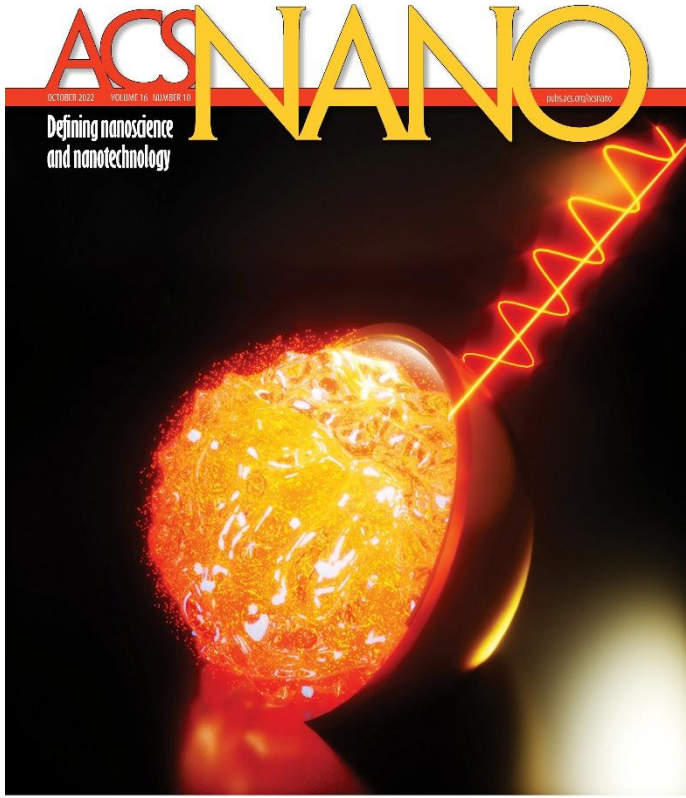
Lensless microscopy: Bypassing lenses entirely through innovative phase modulation of the light

Physics-inspired Deep Learning Instrument Design: PIDL not only reconstruct images, but also helps designing the experimental setup in order to produce data optimized for their processing

Join our Non-Conventional Optics for Life team!

Theoretical statistical Optics
Experimental setups
Data processing and Machine Learning

Applications in Biology, Health and the Environment



INSTITUT POLYTECHNIQUE DE PARIS

ENSTA ENSAE TELECOM Paris TELECOM SudParis